

CRISTOFF STAN

Senior Software Engineer

Location: Tallinn, Harjumaa, Estonia | Email: cristoffstan@gmail.com
LinkedIn: <https://www.linkedin.com/in/cristoff-stan-b53490362/> | Phone Number: +372 58619311

PROFESSIONAL SUMMARY

Accomplished Senior Android Developer with 9+ years of progressive experience in building secure, high-performance mobile applications using Kotlin, Java, and C/C++. Demonstrated expertise in Android internals, low-level system APIs, and secure app architectures. Proven success in implementing MVVM and MVP patterns, optimizing app memory usage, enhancing security with cryptographic practices, and integrating with VPN and secure storage systems. Adept at delivering robust applications in Agile environments and collaborating across distributed teams for large-scale mobile solutions.

TECHNICAL SKILLS

- **Languages:** Kotlin, Java, C, C++, Python, JavaScript
- **Android Expertise:** Jetpack Libraries, VpnService API, NDK, Android Keystore, AOSP internals
- **Security:** Secure Boot, Trusted Execution Environment (TEE), PKI, Keychain, cryptographic protocols
- **Architecture:** MVVM, MVP, Repository Pattern, Clean Architecture
- **Tools:** Android Studio, Git, Firebase, ProGuard, GDB, Valgrind, Wireshark
- **Performance:** Memory leak detection, profiling (Systrace, Traceview), battery & CPU optimization
- **DevOps & Testing:** CI/CD (GitHub Actions, Bitrise), Espresso, JUnit, static code analysis
- **Networking:** REST, Retrofit, WebSockets, SSL pinning
- **Databases:** Room, SQLite, Realm
- **Cloud & Backend:** Firebase, AWS (Cognito, S3), WebRTC, MQTT

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Architected and developed Android applications using Kotlin and C++, implementing secure authentication flows and encrypted data persistence.
- Refactored core components to align with MVVM and Clean Architecture, enhancing scalability and testing capability across codebase.
- Integrated Android's VpnService API for custom VPN tunneling, ensuring privacy-first design with encrypted tunnels and PKI-based handshakes.
- Led Android security initiatives, implementing key obfuscation with ProGuard, encrypted SharedPreferences, and Secure Boot validation.

- Tuned app performance using Android Profiler, cutting down memory usage by 35% through leak analysis and GC optimization.
- Designed a cross-platform C++ layer for core logic, boosting portability and reducing code duplication across mobile platforms.
- Conducted advanced debugging using GDB and Logcat for native crashes, implementing low-level fixes within NDK modules.
- Collaborated in Agile sprints with security engineers and backend teams to meet compliance for financial-grade app standards.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Developed Kotlin-based Android applications for enterprise clients, integrating secure user flows and encrypted offline caching.
- Implemented device-level security checks for rooted/jailbroken device detection using SafetyNet and custom security modules.
- Built a cryptographic signing module using Android Keystore and ECDSA for secure document validation within the app.
- Integrated biometric authentication and secure key handling for high-security mobile solutions in marketing and finance domains.
- Built asynchronous job queues with WorkManager and custom retries for offline-first behavior in unreliable network environments.
- Authored Android modules with strict performance SLAs, including camera-intensive AR applications with real-time processing.
- Created reusable components for user flows using Jetpack Navigation and ViewModels to support modular deployment.
- Conducted quarterly threat modeling reviews and implemented static analysis pipelines to identify security risks early.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Designed custom SDKs in Java/Kotlin and native C++ for partner integration, including VPN tunneling and custom certificate pinning.
- Built a low-level system diagnostic tool leveraging Android's Debug Bridge and profiling APIs to monitor memory and performance.
- Integrated AES-256 and RSA encryption with hybrid key exchange, creating secure end-to-end encrypted messaging channels.
- Optimized legacy Android apps by reducing launch time and improving app responsiveness through view hierarchy flattening.
- Collaborated with system engineers to implement in-app secure logs with tamper detection using SHA hashing techniques.
- Deployed CI pipelines with Bitrise and Fastlane, automating code signing, testing, and rollout to multiple client profiles.
- Developed a custom licensing system tied to device identifiers and secure boot signature validation.
- Spearheaded internal code quality programs including unit test coverage metrics and code review automation.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Developed Android utilities with native C libraries for media streaming, including frame-level video processing using FFmpeg.
- Built proof-of-concept apps to test Android's low-level file system access and interprocess communication limits.
- Reverse-engineered legacy APKs to support modernization and secure system integration with updated APIs.
- Implemented custom shell-level commands through JNI to bridge user apps with system functions.
- Contributed to kernel-level bug tracking via Android Open Source Project submissions.
- Created load tests for background services and optimized wake lock handling to reduce battery impact.
- Drafted Android app architecture blueprints for an in-house secure communications project.
- Interfaced with Linux device drivers for internal performance analytics on test devices.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Assisted in developing a secure Android vault app, using AES encryption, biometric login, and sandboxing principles.
- Conducted threat surface analysis on app permissions and storage access scopes for enhanced user privacy.
- Wrote JNI wrappers for legacy C code to expose secure algorithms within Kotlin layers.
- Performed static analysis and patch reviews for cryptographic libraries used in mobile security modules.
- Created internal tools for APK size reduction through resource shrinking and method tracing.
- Built test automation using UI Automator for security-critical user flows across multiple device configurations.
- Documented memory usage patterns and aligned garbage collection behavior to app lifecycle events.
- Presented research findings on TEE-backed secure operations at internal developer symposiums.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ **Android Security Essentials** – Udacity
- ❖ **Mobile App Security and Reverse Engineering** – Coursera
- ❖ **Kotlin for Android Developers** – JetBrains Academy
- ❖ **Cryptography I** – Stanford Online (Coursera)
- ❖ **Operating Systems and You: Becoming a Power User** – Google IT Support

PROJECTS

- ❖ **SecureVPN Mobile Suite:** Built a zero-logging, encrypted Android VPN app with VpnService API and dynamic tunnel switching.
- ❖ **VaultX:** Designed a secure vault app using Keystore, biometric protection, AES-256 encryption, and hardware-backed security.
- ❖ **ThreatGuard SDK:** Custom Android security SDK providing root detection, anti-tampering, and certificate pinning for 3rd-party apps.
- ❖ **MicroKernel-Analytics:** Developed a lightweight, kernel-level analytics tool for Android using C and log buffers, shipped with secure apps.
- ❖ **NDKProfiler:** Built a performance and memory profiler using C++ and Android NDK to analyze JNI overhead and native memory leaks.